```
mysql>
mysql> CREATE DATABASE lab4;
Query OK, 1 row affected (0.01 sec)

mysql> USE lab4;
Database changed
mysql> CREATE TABLE student(RegNo INT, SName VARCHAR(20), Age INT, Branch VARCHAR(10));
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO student VALUES(1,'Ravi',20,'CSE');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO student VALUES(2,'Kavya',21,'ECE');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student VALUES(3,'Ramesh',22,'CSE');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student VALUES(4,'Manoj',20,'EEE');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student VALUES(5,'Sita',21,'CSE');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM student;
+-------+--------+------+--------+
| RegNo | SName  | Age  | Branch |
+-------+--------+------+--------+
|     1 | Ravi   |   20 | CSE    |
|     2 | Kavya  |   21 | ECE    |
|     3 | Ramesh |   22 | CSE    |
|     4 | Manoj  |   20 | EEE    |
|     5 | Sita   |   21 | CSE    |
+-------+--------+------+--------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM student WHERE Age=20;
+-------+-------+------+--------+
| RegNo | SName | Age  | Branch |
+-------+-------+------+--------+
|     1 | Ravi  |   20 | CSE    |
|     4 | Manoj |   20 | EEE    |
+-------+-------+------+--------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM student WHERE Branch='CSE';
+-------+--------+------+--------+
| RegNo | SName  | Age  | Branch |
+-------+--------+------+--------+
|     1 | Ravi   |   20 | CSE    |
|     3 | Ramesh |   22 | CSE    |
|     5 | Sita   |   21 | CSE    |
+-------+--------+------+--------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student WHERE Age>20;
+-------+--------+------+--------+
| RegNo | SName  | Age  | Branch |
+-------+--------+------+--------+
|     2 | Kavya  |   21 | ECE    |
|     3 | Ramesh |   22 | CSE    |
|     5 | Sita   |   21 | CSE    |
+-------+--------+------+--------+
3 rows in set (0.00 sec)
```

```
CREATE DATABASE lab5' at line 1
mysql> USE lab5;
ERROR 1049 (42000): Unknown database 'lab5'
mysql> CREATE TABLE employee(EmpNo INT, EName VARCHAR(20), Salary INT, DeptNo INT);
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> INSERT INTO employee VALUES(1,'Ravi',30000,10);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO employee VALUES(2,'Kavya',40000,20);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO employee VALUES(3,'Manoj',25000,10);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO employee VALUES(4,'Sita',50000,30);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO employee VALUES(5,'Ramesh',45000,20);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM employee;
+-------+--------+--------+--------+
| EmpNo | EName  | Salary | DeptNo |
+-------+--------+--------+--------+
|     1 | Ravi   |  30000 |     10 |
|     2 | Kavya  |  40000 |     20 |
|     3 | Manoj  |  25000 |     10 |
|     4 | Sita   |  50000 |     30 |
|     5 | Ramesh |  45000 |     20 |
+-------+--------+--------+--------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE Salary BETWEEN 30000 AND 50000;
+-------+--------+--------+--------+
| EmpNo | EName  | Salary | DeptNo |
+-------+--------+--------+--------+
|     1 | Ravi   |  30000 |     10 |
|     2 | Kavya  |  40000 |     20 |
|     4 | Sita   |  50000 |     30 |
|     5 | Ramesh |  45000 |     20 |
+-------+--------+--------+--------+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO student VALUES(2,'Kavya',92);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO student VALUES(3,'Manoj',76);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO student VALUES(4,'Sita',89);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM student;
+------+-------+-------+
| SID  | SName | Marks |
+------+-------+-------+
|    1 | Ravi  |    85 |
|    2 | Kavya |    92 |
|    3 | Manoj |    76 |
|    4 | Sita  |    89 |
|    1 | Ravi  |    85 |
|    2 | Kavya |    92 |
|    3 | Manoj |    76 |
|    4 | Sita  |    89 |
+------+-------+-------+
8 rows in set (0.00 sec)

mysql>
mysql> -- Subquery example
mysql> SELECT SName, Marks FROM student WHERE Marks = (SELECT MAX(Marks) FROM student);
+-------+-------+
| SName | Marks |
+-------+-------+
| Kavya |    92 |
| Kavya |    92 |
+-------+-------+
2 rows in set (0.00 sec)

mysql>
mysql> -- Correlated Query example
mysql> SELECT S1.SName, S1.Marks FROM student S1 WHERE S1.Marks > (SELECT AVG(S2.Marks) FROM student S2);
```

```
mysql> CREATE DATABASE lab6;
Query OK, 1 row affected (0.04 sec)

mysql> USE lab6;
Database changed
mysql> CREATE TABLE sales(SaleID INT, Product VARCHAR(20), Quantity INT, Price INT);
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO sales VALUES(1,'Pen',10,5);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO sales VALUES(2,'Book',5,50);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(3,'Pen',20,5);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(4,'Pencil',15,3);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(5,'Book',10,50);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(6,'Pen',5,5);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM sales;
+--------+---------+----------+-------+
| SaleID | Product | Quantity | Price |
+--------+---------+----------+-------+
|      1 | Pen     |       10 |     5 |
|      2 | Book    |        5 |    50 |
|      3 | Pen     |       20 |     5 |
|      4 | Pencil  |       15 |     3 |
|      5 | Book    |       10 |    50 |
|      6 | Pen     |        5 |     5 |
+--------+---------+----------+-------+
6 rows in set (0.00 sec)

mysql> SELECT Product, SUM(Quantity) FROM sales GROUP BY Product;
+---------+---------------+
| Product | SUM(Quantity) |
+---------+---------------+
| Pen     |            35 |
| Book    |            15 |
| Pencil  |            15 |
+---------+---------------+
```

```
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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> -- =====================================================
mysql> -- Database Creation
mysql> -- =====================================================
mysql> CREATE DATABASE IF NOT EXISTS StringFuncDB;
Query OK, 1 row affected (0.00 sec)

mysql> USE StringFuncDB;
Database changed
mysql>
mysql> -- =====================================================
mysql> -- 1. Create Sample Table
mysql> -- =====================================================
mysql> CREATE TABLE IF NOT EXISTS SampleText (
    ->     ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     TextData VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> -- Insert Sample Data
mysql> INSERT INTO SampleText (TextData) VALUES
    -> ('Hello World'),
    -> ('MySQL String Functions'),
    -> ('Database Management');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- =====================================================
mysql> -- 2. REPLACE Function - Replace a substring
mysql> -- =====================================================
mysql> -- Replace 'World' with 'MySQL' in TextData
mysql> SELECT ID, TextData, REPLACE(TextData, 'World', 'MySQL') AS ReplacedText
    -> FROM SampleText;
+----+------------------------+------------------------+
| ID | TextData               | ReplacedText           |
+----+------------------------+------------------------+
|  1 | Hello World            | Hello MySQL            |
|  2 | MySQL String Functions | MySQL String Functions |
|  3 | Database Management    | Database Management    |
+----+------------------------+------------------------+
3 rows in set (0.00 sec)

mysql>
```

```
Enter password: *********************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.0.43 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> -- ====================================================
mysql> -- Database Creation
mysql> -- ====================================================
mysql> CREATE DATABASE IF NOT EXISTS HLPECursorsDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE HLPECursorsDB;
Database changed
mysql>
mysql> -- ====================================================
mysql> -- 1. Create Sample Table
mysql> -- ====================================================
mysql> CREATE TABLE IF NOT EXISTS Employee (
    ->     EmpID INT PRIMARY KEY AUTO_INCREMENT,
    ->     EmpName VARCHAR(100),
    ->     Department VARCHAR(50),
    ->     Salary DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> -- Insert Sample Data
mysql> INSE
```

```
Enter password: *********************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 8.0.43 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> -- ==========================================================
mysql> -- Database Creation
mysql> -- ==========================================================
mysql> CREATE DATABASE IF NOT EXISTS HLPEFunctionsDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE HLPEFunctionsDB;
Database changed
mysql>
mysql> -- ==========================================================
mysql> -- 1. Simple Function - Return a Constant Value
mysql> -- ==========================================================
mysql> DELIMITER //
mysql> CREATE FUNCTION SimpleFunction()
    -> RETURNS VARCHAR(100)
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN 'Hello! This is a simple function.';
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> -- Call Function
mysql> SELECT SimpleFunction() AS Message;
+-----------------------------------+
| Message                           |
+-----------------------------------+
| Hello! This is a simple function. |
+-----------------------------------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> -- ==========================================================
mysql> -- 2. Function with Input
mysql>
mysql> |
```

```
Enter password: **********************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 39
Server version: 8.0.43 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> -- =====================================================
mysql> -- Database Creation
mysql> -- =====================================================
mysql> CREATE DATABASE IF NOT EXISTS HLPExtensionsDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE HLPExtensionsDB;
Database changed
mysql>
mysql> -- =====================================================
mysql> -- 1. Simple Procedure - No Parameters
mysql> -- =====================================================
mysql> DELIMITER //
mysql> CREATE PROCEDURE SimpleProcedure()
    -> BEGIN
    ->     SELECT 'Hello! This is a simple procedure.' AS Message;
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> -- Call Procedure
mysql> CALL SimpleProcedure();
+------------------------------------+
| Message                            |
+------------------------------------+
| Hello! This is a simple procedure. |
+------------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql>
mysql>
mysql> -- =====================================================
mysql> -- 2. Procedure with Input Parameter
mysql> -- ======================================
mysql> |
```

```
Enter password: *********************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 39
Server version: 8.0.43 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> -- =====================================================
mysql> -- Database Creation
mysql> -- =====================================================
mysql> CREATE DATABASE IF NOT EXISTS HLPExtensionsDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE HLPExtensionsDB;
Database changed
mysql>
mysql> -- =====================================================
mysql> -- 1. Simple Procedure - No Parameters
mysql> -- =====================================================
mysql> DELIMITER //
mysql> CREATE PROCEDURE SimpleProcedure()
    -> BEGIN
    ->     SELECT 'Hello! This is a simple procedure.' AS Message;
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> -- Call Procedure
mysql> CALL SimpleProcedure();
+------------------------------------+
| Message                            |
+------------------------------------+
| Hello! This is a simple procedure. |
+------------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql>
mysql>
mysql> -- =====================================================
mysql> -- 2. Procedure with Input Parameter
mysql> -- ======================================
mysql>
```

```
Enter password: *********************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 37
Server version: 8.0.43 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input stateme

mysql> -- ====================================================
mysql> -- Database Creation
mysql> -- ====================================================
mysql> CREATE DATABASE IF NOT EXISTS CaseLoopDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE CaseLoopDB;
Database changed
mysql>
mysql> -- ====================================================
mysql> -- 1. Simple CASE Example - Grade Calculation
mysql> -- ====================================================
mysql> DELIMITER //
mysql> CREATE PROCEDURE CaseExample1()
    -> BEGIN
    ->     DECLARE marks INT DEFAULT 85;
    ->     DECLARE grade VARCHAR(10);
    ->
    ->     SET grade = CASE
    ->                     WHEN marks >= 90 THEN 'A+'
    ->                     WHEN marks >= 80 THEN 'A'
    ->                     WHEN marks >= 70 THEN 'B'
    ->                     WHEN marks >= 60 THEN 'C'
    ->                     ELSE 'F'
    ->                 END;
    ->
    ->     SELECT marks AS Marks, grade AS Grade;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
```

```
+--------+
1 row in set (0.00 sec)

+--------+
| Number |
+--------+
|      2 |
+--------+
1 row in set (0.00 sec)

+--------+
| Number |
+--------+
|      3 |
+--------+
1 row in set (0.00 sec)

+--------+
| Number |
+--------+
|      4 |
+--------+
1 row in set (0.01 sec)

+--------+
| Number |
+--------+
|      5 |
+--------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql>
mysql>
mysql> -- ==================================================
mysql> -- 3. REPEAT Loop - Sum of First 5 Numbers
mysql> -- ==================================================
mysql> DELIMITER //
mysql> CREATE PROCEDURE RepeatExample2()
    -> BEGIN
    ->     DECLARE counter IN
    ->
```

```
mysql>
mysql> CREATE DATABASE labdb2;
ERROR 1007 (HY000): Can't create database 'labdb2'; database exists
mysql>
mysql> USE labdb2;
Database changed
mysql>
mysql> CREATE TABLE department (DeptNo INT PRIMARY KEY, DName VARCHAR(20) UNIQUE, Location VARCHAR(15));
ERROR 1050 (42S01): Table 'department' already exists
mysql>
mysql> DESC department;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| DeptNo   | int         | NO   | PRI | NULL    |       |
| DName    | varchar(20) | YES  | UNI | NULL    |       |
| Location | varchar(15) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE faculty (FacNo INT PRIMARY KEY, FName VARCHAR(20), Gender CHAR(1) CHECK (Gender IN ('M','F')), DeptNo INT, FOREIGN KEY (DeptNo) REFERENC
ES department(DeptNo));
ERROR 1050 (42S01): Table 'faculty' already exists
mysql>
mysql> DESC faculty;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| FacNo  | int         | NO   | PRI | NULL    |       |
| FName  | varchar(20) | YES  |     | NULL    |       |
| Gender | char(1)     | YES  |     | NULL    |       |
| DeptNo | int         | YES  | MUL | NULL    |       |
+--------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

```
mysql>
mysql> -- (b) Create a View for Department-wise Average Salary
mysql> CREATE VIEW Dept_AvgSalary AS
    -> SELECT Department, AVG(Salary) AS Avg_Salary
    -> FROM Employee
    -> GROUP BY Department;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> -- Use the View
mysql> SELECT * FROM Dept_AvgSalary;
+------------+--------------+
| Department | Avg_Salary   |
+------------+--------------+
| HR         | 45000.000000 |
| IT         | 67500.000000 |
| Finance    | 60000.000000 |
+------------+--------------+
3 rows in set (0.00 sec)

mysql>
mysql> -- =========================
mysql> -- 5. INDEX Examples
mysql> -- =========================
mysql>
mysql> -- (a) Create an Index on Department column
mysql> CREATE INDEX idx_department ON Employee(Department);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> -- (b) Create a Unique Index on EmpName
mysql> CREATE UNIQUE INDEX idx_empname ON Employee(EmpName);
ERROR 1062 (23000): Duplicate entry 'Alice Brown' for key 'employee.idx_empname'
mysql>
mysql> -- Check Indexes
mysql> SHOW INDEX FROM Employee;
+----------+------------+----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table    | Non_unique | Key_name       | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+----------+------------+----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| employee |          0 | PRIMARY        |            1 | EmpID       | A         |          12 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| employee |          1 | idx_department |            1 | Department  | A         |           3 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+----------+------------+----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
2 rows in set (0.02 sec)

mysql>
mysql> -- (c) Use Indexed Query
mysql> SELECT * FROM Employee WHERE Department = 'IT';
+-------+-------------+------------+----------+
| EmpID | EmpName     | Department | Salary   |
+-------+-------------+------------+----------+
|     2 | Alice Brown | IT         | 60000.00 |
|     4 | Emma Davis  | IT         | 75000.00 |
|     8 | Alice Brown | IT         | 60000.00 |
|    10 | Emma Davis  | IT         | 75000.00 |
+-------+-------------+------------+----------+
4 rows in set (0.00 sec)
```

```
>> ('Headphones', 'Electronics', 10, 1500, '2025-09-21'),
>> ('Chair', 'Furniture', 7, 2000, '2025-09-22'),
>> ('Table', 'Furniture', 4, 3000, '2025-09-23'),
>> ('Mouse', 'Electronics', 15, 800, '2025-09-24'),
>> ('Sofa', 'Furniture', 2, 15000, '2025-09-25');
Laptop
Electronics
5
50000
2025-09-20
Headphones
Electronics
10
1500
2025-09-21
Chair
Furniture
7
2000
2025-09-22
Table
Furniture
4
3000
2025-09-23
Mouse
Electronics
15
800
2025-09-24
Sofa
Furniture
2
15000
2025-09-25
PS C:\Users\vashi>
```

```
mysql>
mysql> CREATE DATABASE lab3;
Query OK, 1 row affected (0.01 sec)

mysql> USE lab3;
Database changed
mysql> CREATE TABLE department(deptno INT, dname VARCHAR(20), hod VARCHAR(20));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE faculty(facno INT, fname VARCHAR(20), gender CHAR(1), mobileno BIGINT, doj DATE, deptno INT);
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE student2(RegNo INT, SIName VARCHAR(20), Age INT, MobileNo BIGINT, Facno INT);
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO department VALUES(1,'CSE','Ravi');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO department VALUES(2,'ECE','Sita');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM department;
+--------+-------+------+
| deptno | dname | hod  |
+--------+-------+------+
|      1 | CSE   | Ravi |
|      2 | ECE   | Sita |
+--------+-------+------+
2 rows in set (0.00 sec)

mysql>
mysql> INSERT INTO faculty VALUES(101,'Anil','M',9876543210,'2020-06-01',1);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO faculty VALUES(102,'Kavya','F',9876501234,'2021-01-10',2);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM faculty;
+-------+-------+--------+------------+------------+--------+
| facno | fname | gender | mobileno   | doj        | deptno |
+-------+-------+--------+------------+------------+--------+
|   101 | Anil  | M      | 9876543210 | 2020-06-01 |      1 |
|   102 | Kavya | F      | 9876501234 | 2021-01-10 |      2 |
+-------+-------+--------+------------+------------+--------+
2 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE DATABASE lab4;
Query OK, 1 row affected (0.01 sec)

mysql> USE lab4;
Database changed
mysql> CREATE TABLE student(RegNo INT, SName VARCHAR(20), Age INT, Branch VARCHAR(10));
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO student VALUES(1,'Ravi',20,'CSE');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO student VALUES(2,'Kavya',21,'ECE');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student VALUES(3,'Ramesh',22,'CSE');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student VALUES(4,'Manoj',20,'EEE');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO student VALUES(5,'Sita',21,'CSE');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM student;
+-------+--------+------+--------+
| RegNo | SName  | Age  | Branch |
+-------+--------+------+--------+
|     1 | Ravi   |   20 | CSE    |
|     2 | Kavya  |   21 | ECE    |
|     3 | Ramesh |   22 | CSE    |
|     4 | Manoj  |   20 | EEE    |
|     5 | Sita   |   21 | CSE    |
+-------+--------+------+--------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM student WHERE Age=20;
+-------+-------+------+--------+
| RegNo | SName | Age  | Branch |
+-------+-------+------+--------+
|     1 | Ravi  |   20 | CSE    |
|     4 | Manoj |   20 | EEE    |
+-------+-------+------+--------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM student WHERE Branch='CSE';
+-------+--------+------+--------+
| RegNo | SName  | Age  | Branch |
+-------+--------+------+--------+
|     1 | Ravi   |   20 | CSE    |
|     3 | Ramesh |   22 | CSE    |
|     5 | Sita   |   21 | CSE    |
+-------+--------+------+--------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student WHERE Age>20;
+-------+--------+------+--------+
| RegNo | SName  | Age  | Branch |
+-------+--------+------+--------+
|     2 | Kavya  |   21 | ECE    |
|     3 | Ramesh |   22 | CSE    |
|     5 | Sita   |   21 | CSE    |
+-------+--------+------+--------+
3 rows in set (0.00 sec)
```

```
mysql> CREATE DATABASE lab6;
Query OK, 1 row affected (0.04 sec)

mysql> USE lab6;
Database changed
mysql> CREATE TABLE sales(SaleID INT, Product VARCHAR(20), Quantity INT, Price INT);
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO sales VALUES(1,'Pen',10,5);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO sales VALUES(2,'Book',5,50);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(3,'Pen',20,5);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(4,'Pencil',15,3);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(5,'Book',10,50);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO sales VALUES(6,'Pen',5,5);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM sales;
+--------+---------+----------+-------+
| SaleID | Product | Quantity | Price |
+--------+---------+----------+-------+
|      1 | Pen     |       10 |     5 |
|      2 | Book    |        5 |    50 |
|      3 | Pen     |       20 |     5 |
|      4 | Pencil  |       15 |     3 |
|      5 | Book    |       10 |    50 |
|      6 | Pen     |        5 |     5 |
+--------+---------+----------+-------+
6 rows in set (0.00 sec)

mysql> SELECT Product, SUM(Quantity) FROM sales GROUP BY Product;
+---------+---------------+
| Product | SUM(Quantity) |
+---------+---------------+
| Pen     |            35 |
| Book    |            15 |
| Pencil  |            15 |
+---------+---------------+
```

```
mysql> INSERT INTO student VALUES(2,'Kavya',92);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO student VALUES(3,'Manoj',76);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO student VALUES(4,'Sita',89);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM student;
+------+-------+-------+
| SID  | SName | Marks |
+------+-------+-------+
|    1 | Ravi  |    85 |
|    2 | Kavya |    92 |
|    3 | Manoj |    76 |
|    4 | Sita  |    89 |
|    1 | Ravi  |    85 |
|    2 | Kavya |    92 |
|    3 | Manoj |    76 |
|    4 | Sita  |    89 |
+------+-------+-------+
8 rows in set (0.00 sec)

mysql>
mysql> -- Subquery example
mysql> SELECT SName, Marks FROM student WHERE Marks = (SELECT MAX(Marks) FROM student);
+-------+-------+
| SName | Marks |
+-------+-------+
| Kavya |    92 |
| Kavya |    92 |
+-------+-------+
2 rows in set (0.00 sec)

mysql>
mysql> -- Correlated Query example
mysql> SELECT S1.SName, S1.Marks FROM student S1 WHERE S1.Marks > (SELECT AVG(S2.Marks) FROM student S2);
```

```
CREATE DATABASE lab5' at line 1
mysql> USE lab5;
ERROR 1049 (42000): Unknown database 'lab5'
mysql> CREATE TABLE employee(EmpNo INT, EName VARCHAR(20), Salary INT, DeptNo INT);
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> INSERT INTO employee VALUES(1,'Ravi',30000,10);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO employee VALUES(2,'Kavya',40000,20);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO employee VALUES(3,'Manoj',25000,10);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO employee VALUES(4,'Sita',50000,30);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO employee VALUES(5,'Ramesh',45000,20);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM employee;
+-------+--------+--------+--------+
| EmpNo | EName  | Salary | DeptNo |
+-------+--------+--------+--------+
|     1 | Ravi   |  30000 |     10 |
|     2 | Kavya  |  40000 |     20 |
|     3 | Manoj  |  25000 |     10 |
|     4 | Sita   |  50000 |     30 |
|     5 | Ramesh |  45000 |     20 |
+-------+--------+--------+--------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM employee WHERE Salary BETWEEN 30000 AND 50000;
+-------+--------+--------+--------+
| EmpNo | EName  | Salary | DeptNo |
+-------+--------+--------+--------+
|     1 | Ravi   |  30000 |     10 |
|     2 | Kavya  |  40000 |     20 |
|     4 | Sita   |  50000 |     30 |
|     5 | Ramesh |  45000 |     20 |
+-------+--------+--------+--------+
4 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE TABLE faculty (Facno INT, FName CHAR(15), gender CHAR(1), DOB DATE, MobileNo INT, address VARCHAR(20));
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> ALTER TABLE faculty ADD PRIMARY KEY(Facno);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> ALTER TABLE faculty ADD CHECK (gender IN ('M','F'));
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> DESC faculty;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| Facno    | int         | NO   | PRI | NULL    |       |
| FName    | char(15)    | YES  |     | NULL    |       |
| gender   | char(1)     | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| MobileNo | int         | YES  |     | NULL    |       |
| address  | varchar(20) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE department (deptno INT, dname CHAR(15), loc CHAR(10));
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> ALTER TABLE department ADD PRIMARY KEY(deptno);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0
```